The HSAT2 Hypothesis: Viral Centromere Disruption, Epigenetic Erosion, and Immune Exhaustion in Post-Acute Infection Syndromes

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Post-acute infection syndromes (PAIS)—including ME/CFS and Long COVID—lack an established molecular mechanism. I examine a hypothesis centred on human satellite 2 (HSAT2), a repetitive DNA sequence making up ∼1.5% of the human genome, normally kept silent by DNA methylation and tightly packed protein structures. In cancer, HSAT2 RNA escapes this silencing and is exported in extracellular vesicles that suppress the immune system. A structurally analogous cascade is proposed for PAIS: viral disruption of centromeres triggers HSAT2 activation → loss of silencing marks → immune-suppressing particles spread through the body → self-reinforcing immune exhaustion. I evaluate the evidence across four criteria and identify five testable predictions. No step has been demonstrated in PAIS, but the hypothesis's primary strength is falsifiability: every prediction is testable with existing methods and achievable cohort sizes.

1 Background

PAIS affect millions with chronic fatigue, post-exertional malaise, immune dysfunction, and neurocognitive symptoms persisting years after infection [1, 2]. No molecular mechanism is established, and validated biomarkers remain absent. I propose the following chain linking viral triggers to chronic immune pathology: viral disruption of centromeres (the constricted central regions of chromosomes) → HSAT2 RNA transcription → loss of CpG methylation (the chemical silencer mark) → spread via extracellular vesicles → immune exhaustion. The hypothesis draws on cancer biology, virology, and epigenetics but is untested in PAIS. A task force led by Claire Vourc'h (IAB, Université Grenoble Alpes) is forming to test it experimentally.

2 The Evidence Chain

2.1 Trigger: Viral Disruption of Centromeres

Lahaye et al. (2025) showed herpesvirus proteins cause centromeres—the central constrictions of chromosomes—to amplify their DNA [3]. This restructuring is proposed as the trigger for HSAT2 activation, which Patrick Lomonte and I discuss as the likely proximate cause of pericentromeric satellite transcription (pers. comm., May 2026). Nogalski et al. (2019) confirmed that HSV-1 and HCMV switch on HSAT2 RNA in human cells [4]. Target cells include astrocytes, microglia, blood-vessel lining cells, pericytes, and neurons—brain regions targeted by HSV-1, HHV-6, and SARS-CoV-2 overlap substantially, suggesting cooperative infection dynamics that could explain herpesvirus reactivation in Long COVID (P. Lomonte, pers. comm., May 2026).

2.2 Activation: How HSAT2 Gets Switched On

Four non-exclusive pathways may converge to derepress HSAT2:

Oxidative stress. Excess reactive oxygen species activate HSF1, a protein that drives satellite transcription—well-studied for the related repeat HSAT3 [5], less established for HSAT2.

Loss of CTCF. CTCF is a protein that acts as a molecular gatekeeper, maintaining satellite repeat silencing [6, 7]. HSAT2 RNA forms nuclear bodies that trap MeCP2 and CTCF, and separately traps the KDM2A/B-PRC1 silencing complex [8, 9]. Inflammation reduces CTCF binding at these regions [10]. Whether CTCF loss opens HSAT2 in PAIS is untested.

DNA hypomethylation. Altered methylation patterns are documented in ME/CFS [11–13]. Depletion of MTHFR, an enzyme in the methylation cycle, weakens silencing marks at pericentromeric repeats [14].

Transcription factor sequestering. HSAT2 RNA contains GGAA repeats that bind ETS-family transcription factors—proteins that control gene activity. In Ewing sarcoma, aberrant ETS proteins produce pericentromeric RNAs exported in extracellular vesicles [15]. The same repeats also bind SPI1 (PU.1), a master regulator of blood and immune cell development. HSAT2-mediated SPI1 trapping could drive expansion of immune-suppressing cells and NK/T-cell dysfunction.

2.3 Propagation: Immune-Suppressing Particles

Ruzanov et al. (2024) reported that in Ewing sarcoma, HSAT2 and related RNAs are packaged into extracellular vesicles—tiny cell-derived particles that travel through body fluids—which drive expansion of immune-suppressing myeloid cells (MDSCs) and exhaust $CD8^+$ killer T-cells [15]. Recipient cells propagated HSAT2 RNA and released it in their own vesicles, enabling serial spread; AZT reduced accumulation. In osteosarcoma, HSAT2 DNA in blood-derived vesicles achieved near-perfect diagnostic accuracy (AUC = 0.97) [16]. The proposed parallel: HSAT2-containing vesicles from cells with viral centromere disruption reach the bloodstream, target immature myeloid cells, and drive immune suppression—contributing to the NK dysfunction and T-cell exhaustion seen in ME/CFS and Long COVID.

2.4 Consolidation: A Self-Reinforcing Feedback Loop

In cancer, Bonnet, Fourel et al. (2026, preprint) describe how the methylation enzyme DNMT3B is pulled away from repetitive regions (including HSAT2) toward actively transcribed genes and Alu repeats, eroding the silencing marks that normally keep repeats off [17]. For PAIS, causality is reversed: HSAT2 transcription triggers DNMT3B loss as a consequence, not a cause, producing demethylation that makes further transcription easier. The rare genetic disease ICF syndrome (caused by DNMT3B mutations) links pericentromeric demethylation to immune dysfunction [18]. Loss of methylation does not guarantee HSAT2 activation: backup silencing systems—including H1 linker histones, HP1α phase-separated heterochromatin, nucleosome repositioning, and Polycomb invasion—provide redundancy [17]. Whether demethylation drives transcription or vice versa remains experimentally unresolved.

3 Clinical and Translational Implications

Treatment response. Methyl donors (SAMe 1600–3200 mg/d, methylfolate, betaine) may restore silencing by replenishing the raw material for DNA methylation; betaine provides an alternative pathway via BHMT. Riboflavin (vitamin B2) aids methylfolate production in MTHFR C677T carriers; zinc is a structural cofactor for methylation enzymes. NAC, an antioxidant, may reduce satellite transcription by lowering oxidative stress. If the consolidation model is correct, drugs that strip methylation marks (HDAC inhibitors, 5-azacitidine) would worsen the disease.

Patient stratification. HSAT2 activation is expected in a subgroup characterised by documented viral trigger, elevated HSAT2 in blood-derived vesicles, MDSC expansion, and NK dysfunction.

Biomarker detection. Two HSAT2 detection tests exist, both exceeding 90% accuracy in cancer [19, 20]. Validated primer sets are available from Nogalski et al. (2019) and Ting et al. (2011). Sensitivity at the lower concentrations expected in PAIS is unresolved.

4 Falsifiable Predictions

(1) **Plasma EV HSAT2 elevation.** HSAT2 in blood-derived vesicles elevated in a PAIS subgroup with viral trigger.

(2) **Post-exertional HSAT2 increase.** HSAT2 increases 24–48 h after exercise testing, correlating with symptom worsening. *Safety note:* exercise testing carries exacerbation risk; stopping criteria and exclusion of severely affected patients are mandatory.

(3) **MDSC co-clustering.** Immune-suppressing cell frequency and vesicle HSAT2 co-cluster in patients but not controls.

(4) **Methyl donor response.** In patients with elevated baseline HSAT2, methyl donors (SAMe + methylfolate) reduce HSAT2 more than placebo at 12 weeks.

(5) **RT inhibitor effect.** Reverse transcriptase inhibitors (such as AZT) reduce HSAT2 load. *Requires preclinical safety validation.*

All predictions are testable with existing methods.

5 Limitations

No step has been directly demonstrated in PAIS. The evidence comes from a single Ewing sarcoma study [15], an osteosarcoma biomarker study [16], and a 2026 preprint [17]. EBV and HHV-6 have not been tested for HSAT2 induction. Alternative explanations include: cell-type shifts rather than locus-specific changes; HSAT2 activation as a marker rather than driver of disease; and the HSF1 pathway being well-characterised for HSAT3 but not HSAT2. The hypothesis may be disproven upon testing. Its primary contribution is falsifiability: each step generates a testable prediction whose outcome would constrain or refute it.

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